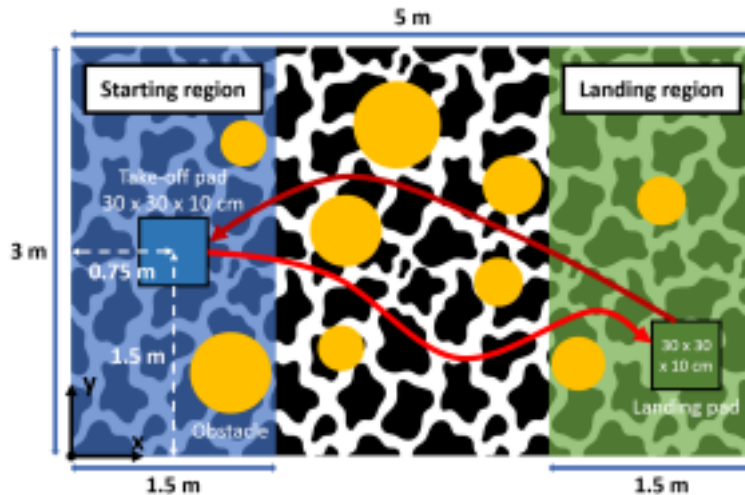


CrazyPracticals

In this practical, you will learn how to program a Crazyflie to find and precisely land on a platform with the help of minimal sensory information. Additionally, you will use sensor readings to design a robust algorithm that can avoid obstacles present in the environment. You will do so in a team of two or three members and show its performance in a live demonstration towards the end of the course.



The arena for the final demonstration is shown in the figure above (dimensions are approximate). The **task** you are required to complete in this arena is composed of multiple phases:

- The UAV takes off from a take-off pad placed within the starting region.
- Thanks to the optic-flow sensor and a z-range finder the UAV can track its movements and find the landing pad in the landing zone.
- Once the landing pad is found, the UAV has to precisely land on it.
- In the workspace, there can be a few obstacles of previously unknown dimensions which it can detect with the multiranger deck. Your developed algorithm should allow the UAV to avoid any collisions.

The **learning objectives** for this practical are:

- assembling/modifying a plug-and-play UAV
- interfacing with the python lib to control the UAV
- mastering different flight phases
- designing a robust algorithm to conduct a flight mission in the presence of obstacles
- reporting performance results in a scientific manner

Some class time will be set aside for this project. Classes on September 22nd, October 6th, and October 20th will be dedicated to flight testing.

After the final flight, you are expected to submit:

- Short video including recording of one of your trials
- 3 trials with obstacles (best grade counts)

The presentation should be presented in the form of a video (max 7 min) and include the recording of one of your trials. A simple way to do this is for everyone to hop on a Zoom call, someone shares screen and you all present. Creativity is encouraged though and other video styles are accepted so long as the

information is conveyed. A Google Drive will be shared for you to upload your video so that everyone can check it out (.mp4 file type). The video should contain:

- The strategy (algorithm, what you spend most time on)
- Your flight test plans (cards, different flights, etc.)
- The results (statistics on mission time/success/...)
- Anything else that is interesting/relevant

The video you submit should show a run demonstrating your algorithm's performance. The video of the run should:

- be max 2 min long
- contain no edits except speed ups (indicate speed up factor) and text additions
- show at least one of the team members

Complementary information

- **Submit by Friday 24th October, 23:59** your code with comments in a zip file on Brightspace.
- Return the material directly after the demonstration.
- Check that the returned material is complete.

The **grading** will be according to the level tasks you are able to solve:

Points	Accomplished Tasks
5	Return to original take off pad.
5	Locate and land on the landing pad
5	Avoid obstacles in the way
5	Take off and move according to a search pattern
10	Presentation and video of working algorithm

*Partial credit is possible if you are unable to complete each task. How much partial credit depends on how close you are to completing it and **most importantly**, how clear and detailed your explanation of what went wrong and how you attempted to debug it/could debug it in the future.*